

DELCODE | PET data documentation

Authors: Marcel Daamen + Henning Boecker

E-Mail: PET-group@dzne.de

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1. Purpose of this document

This document describes the PET examinations of **cerebral glucose metabolism** (using ^{18}F -Fluorodesoxyglucose [FDG]), and of **brain amyloid burden** (using NeuraceqTM [18F-Florbetaben: FBB]: Life Radiopharma, Berlin), respectively, that were performed in the DELCODE study.

The document provides details about both the **image data** and **data analyses** that can be requested by interested researchers.

On the one hand, these descriptions should assist researchers in judging whether the available PET data are suitable for answering their given research question. On the other hand, they provide technical details for the Methods sections in future publications.

Note for publications using DELCODE-PET Florbetaben PET data: The qualitative analyses and the initial quantitative analyses were performed under a research agreement with Life Molecular Imaging. Moreover, the final Florbetaben PET follow-up examinations were performed with financial support from the AMYPAD research consortium. This should be reported in the Acknowledgments/ Sponsoring/ Conflicts of Interest, where appropriate.

2. General study design

2.1. Study population

In general, it must be emphasized that PET examinations were **only collected from Subjective Cognitive Decline (SCD) participants from the DELCODE study** who agreed to these additional examinations.

No complementary PET data are available from healthy controls, MCI and AD dementia participants!

2.2. Imaging modalities (FBB, FDG) and sessions (P0, P24)

DELCODE-PET includes PET examinations of **cerebral glucose metabolism** using ^{18}F -Fluorodesoxyglucose (FDG), and **brain amyloid burden** using Neuraceq™ (18F-Florbetaben [FBB]; Life Radiopharma, Berlin).

For both FBB and FDG, PET examinations were **performed up to two times**: A baseline scan ("P0") and a follow-up scan ("P24"). Accordingly, the sessions are labelled with combinations of tracer and timepoint (e.g., FBB_P24).

Note: Due to the delayed start of DELCODE-PET, the baseline scan was not always performed at the baseline visit of the DELCODE main study, but during follow-up visits of the DELCODE main study. Accordingly, the "P0" and "P24" labels for the PET data are not equivalent with the "M0, M12, M24, ..." labels found in the MRI data. Moreover, while follow-up PET scans were originally scheduled for a 24-month delay (hence: "P24"), this was later extended to a time window of up to 36 months.

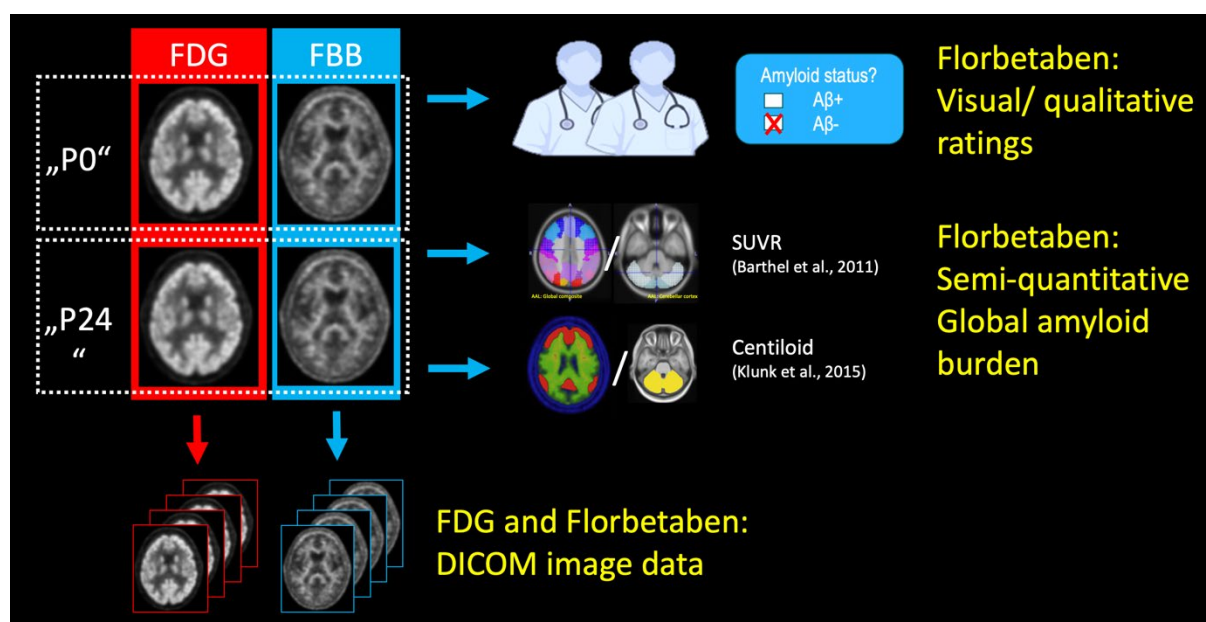


Figure 1. DELCODE-PET: Available data summary

2.3. Data acquisition

PET data acquisition followed established standard protocols. While there were tracer-specific differences in preparation and tracer application, PET image acquisition was technically similar:

	FDG	FBB
Requested time since last food intake	≥ 6 hours	Not applicable
Maximum blood glucose level	≤ 140 ng/ml	Not applicable
Applied tracer activity	170-180 Megabequerel (MBq)	260-300 Megabequerel (MBq)
Rest period after injection	20 min	Not applicable
Start of emission scan	40 min post-injection	90 min post-injection
Duration of emission	20 min	
Acquisition mode	3D Listmode, <i>post hoc</i> reconstruction in 4*5 min time frames	
Reconstruction algorithm	<i>Iterative:</i> Based on established brain protocols at PET study sites • <i>Siemens + GE:</i> Ordered Subset Expectation Maximization (OSEM) • <i>Philips:</i> Line-of-response row-action maximum likelihood (LOR-RAMLA)	
Image corrections	• Decay • Randoms • Dead time • Scatter • Attenuation • Normalization	
Attenuation correction	Based on vendor-specific attenuation coefficient (μ) maps • <i>Siemens/ GE/ Philips PET/CT systems:</i> Derived from low-dose CT • <i>Siemens PET/MR systems:</i> Derived from DIXON-VIBE MR sequence	

2.4. Quality assessments

After image data transfer, PET images were inspected by the PET Group at the DZNE Bonn to check compliance with the acquisition protocol. This included:

- Check for completeness of the relevant image series
- Extraction of information from the reconstructed PET DICOM series
- Check of consistency of the available information with standard acquisition protocol
- Visual image check for complete brain coverage (including the whole cerebellum)
- Visual image check for image artifacts
- Evaluation of participant motion:
 - An automated estimation of geometric translations and rotations was performed with the *Motion Correction* of the Fusion module of PMOD (PMOD Technologies LLC, Zurich), in order to identify time frames with significant frame displacement (> 4 mm: *QIBA Amyloid PET Profile*: [QIBA, 2022](#)).

In these rare cases, it was checked whether the time frames could be censored, affording that ≥ 2 valid time frames (= a minimum scan time of 10 min) were left for analyses.

3. Which sorts of PET data are available?

Depending on the research question, and individual experience with PET data, the following data types can be requested.

1. **FBB and FDG DICOM images:** The attenuation-corrected iterative reconstructions of the PET emission scans are made available for voxel-based analyses, or custom volume-of-interest (VOI) based analysis pipelines. In addition to the original image series, four additional image series with varying levels of preprocessing are available (Paragraph 4).
2. **FBB visual ratings:** As the most basic analysis of the amyloid PET scans, qualitative ratings of amyloid positivity or negativity can be requested, based on the consensual judgment of two experienced readers (Paragraph 5).
3. **FBB semi-quantitative analyses:** In addition to older *Standardized Uptake Value Ratio* (SUVR) analyses of cortical amyloid burden which were based on the original Florbetaben validation study ([Barthel et al. 2011](#)), we recently added *Centiloid* ([Klunk et al., 2015](#)) values (Paragraph 6).

Currently, no complementary semi-quantitative analyses for the FDG data are provided, but are planned to become available in the future.

4. DICOM image data

In addition to the original, attenuation-corrected PET image series, **four preprocessed versions of this dataset** are provided. While many users will probably prefer the final preprocessed version, there is the option to step back to earlier processing steps, or to the “raw” original reconstruction, depending on the specific research question.

This data preparation strategy was adopted from the ADNI PET preprocessing pipeline (for detailed descriptions: [Jagust et al., 2015](https://adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/); <https://adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/>).

Yet, since the in-house processing pipeline of the ADNI consortium was not available, our data processing in PMOD (Figure 2) required some technical changes, which are explained below:

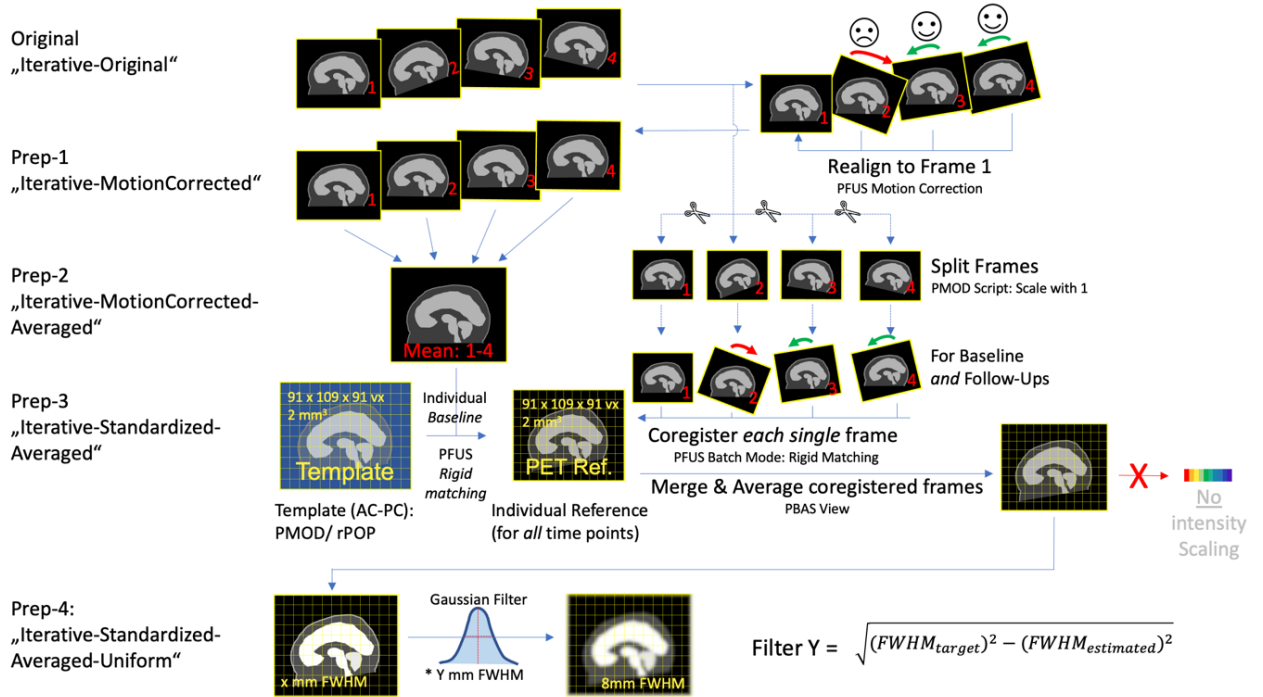


Figure 2. Schematic outline of PET data preprocessing for DELCODE

4.1. Realignment

As an initial preprocessing step, time frames #2 to #4 of the original time series were realigned with reference time frame #1 using the PMOD Fusion Module, and resliced into a new time series.

4.2. Standardization of image geometry

Due to technical differences between the scanner vendors and reconstruction protocols, the voxel geometry and size of the PET DICOM series varied. This is different from the DELCODE MR imaging data, where the uniform use of Siemens MR scanners allowed for a better harmonization of data acquisition protocols.

Similar to the image processing pipeline in ADNI, the PET image data were coregistered to **modality-specific templates**, and subsequently resliced, resulting in images with a uniform 91 x 109 x 91 voxel grid, and 2 x 2 x 2 mm voxel size.

- 1) For each subject, the motion-corrected, averaged image from the baseline PET scan was rigidly matched to the respective modality-specific template (with optional manual corrections after visual inspection), and resliced into template voxel space to serve as a subject-specific reference.
- 2) Each single time frame of *both* baseline *and* follow-up PET scans was rigidly matched to this subject-specific reference, and resliced. As a result, each single time frame is only interpolated once, and should have a similar orientation and voxel space for a given subject, not only *within*, but also *across* baseline and follow-up scans for the respective tracer.
- 3) After visual checking, the coregistered frames of each PET time series were averaged.

Because the original ADNI templates were not available, images were coregistered to a PMOD-custom FDG template (constructed from N=102 ADNI controls), or an FBB-specific template taken from the *rPOP* toolbox ([Iaccarino et al., 2022](#): constructed from N=50 amyloid-positive and N=50 amyloid-negative subjects with FBB scans), respectively.

Note that both coregistration steps (which were optimized by Normalized Mutual Information) applied *rigid body transformations* only.

Similar to ADNI, **no spatial normalization** (e.g., to MNI template space) using affine or nonlinear warping was performed. Moreover, although the standardized images will often be roughly aligned in our personal experience, **no explicit orientation to MNI coordinate space** (e.g., setting of image origin to anterior commissure) is provided.

Finally, be aware that the standardized PET series are **not intensity scaled** (= not “SUVR” images). The respective procedures in ADNI (iterative scaling of global signal mean for FDG, scaling by cerebellar gray matter average for amyloid PET) could not be reproduced satisfactorily in PMOD.

This means that users have to apply their own scaling during image data analysis. This can be achieved by explicit image scaling (e.g., dividing each voxel intensity by the mean signal of a reference region), or implicitly in statistical modelling (e.g., SPM proportional scaling).

4.3. Estimation of PET scanner/ reconstruction resolution

Independent from voxel geometry, PET reconstructions differ regarding their effective spatial resolution (based on scanner-specific factors like detector technology, reconstruction method and post-reconstruction filter).

We adopted a method introduced in [Joshi et al. \(2009\)](#) to estimate spatial resolution empirically, based on 3D-Hoffman brain phantom scans with the acquisition protocols that were collected at the study sites. Basically, the unknown resolution was inferred by applying Gaussian smoothing to a digital version of the Hoffman phantom to identify the smoothing parameter settings that produced the least mismatch between the *measured* phantom and the smoothed *digital* phantom (see Figure 3).

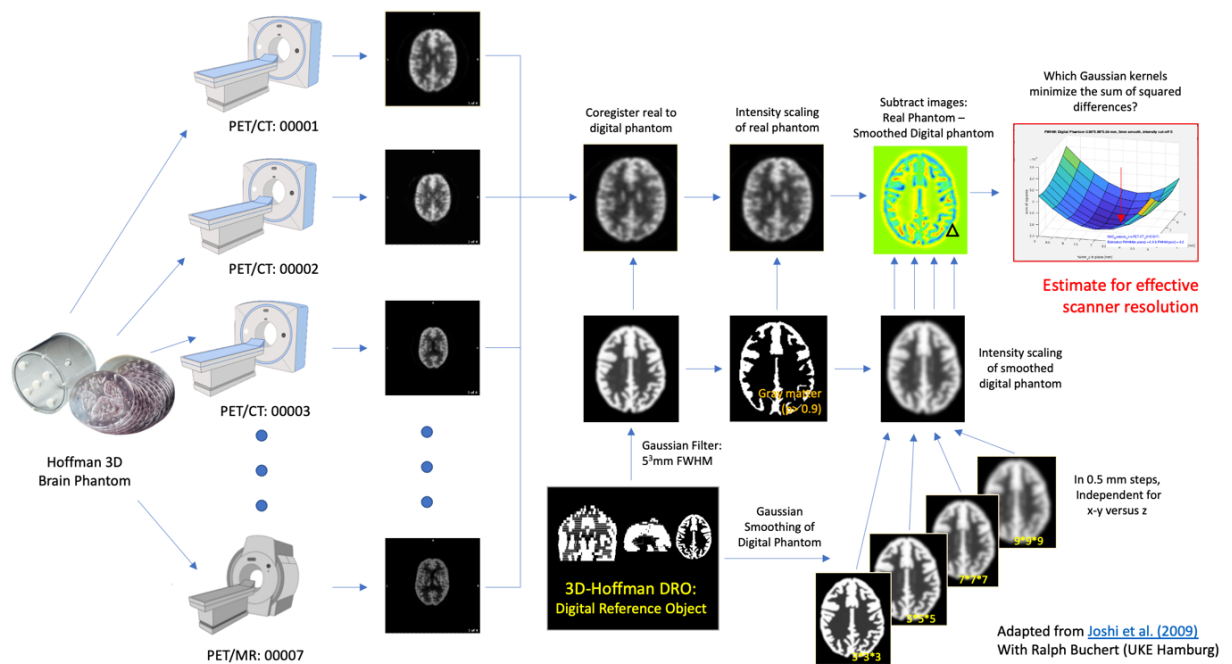


Figure 3. Estimation of the intrinsic scanner/ reconstruction resolutions

The procedure was implemented in a Matlab script originally written by Ralph Buchert (Universitätsklinik Eppendorf, Hamburg), with slight modifications. It adapted the processing steps described in the *QIBA Amyloid PET Profile* ([QIBA, 2022: Appendix H](#)). Because the original digital phantom was not available, a more recent, publicly available version with a higher voxel resolution was used ([Harrison et al., 2020](#)). Smoothing kernels were varied independently for the x-y and z dimensions, in 0.5 mm steps, to find the solution where subtraction of the two coregistered images resulted in minimum sum of error squares.

Based on these estimates, the size of the Gaussian filter (in x-y versus z direction) that was necessary for a target smoothness of 8 mm FWHM was approximated, with:

$$\text{Filter} = \sqrt{(FWHM_{\text{target}})^2 - (FWHM_{\text{estimated}})^2}$$

Again, filter sizes were rounded to the closest 0.5 mm value.

The chosen target resolution of 8mm FWHM is compatible with ADNI, but was also the least common denominator necessary for including the scanner with the lowest resolution. *In principle*, you could drop the latter, take the unsmoothed standardized images from Prep-3, and use the formula to calculate Gaussian kernels that are necessary to smooth these PET images to an alternative resolution (e.g., using SPM).

Yet, note that the majority of the available PET data had reconstructions with estimated 7 mm resolution, leaving little space for improvement without sacrificing substantial amounts of data.

4.4. Available image versions

To summarize, a total of 5 image series are provided per PET examination:

Series description	Details
Original: <i>"Iterative-Original"</i>	Time series with 4 frames of 5 min duration each. The attenuation-corrected, iterative reconstruction of the original PET acquisition, as uploaded by the PET sites, retaining the individual scanner-specific image geometry (i.e., reconstruction matrix, in-plane pixel size, slice thickness and number) and effective image resolution.
Prep-1: <i>"Iterative-MotionCorrected"</i>	Time series with 4 frames of 5 min duration each. After realignment of time frames #2 -4 with frame #1. The original scanner-specific image geometry and effective image resolution is preserved.
Prep-2: <i>"Iterative-MotionCorrected-Averaged"</i>	Single time frame of 20 min duration (unless censored). Average of the motion-corrected series from Prep-1. The original scanner-specific image geometry and effective image resolution is preserved.
Prep-3: <i>"Iterative-Standardized-Averaged"</i>	Single time frame of 20 min duration (unless censored). Interpolated PET average with a uniform voxel geometry (91 x 109 x 91 voxel grid, and 2 x 2 x 2 mm voxel size).

Prep-4: <i>"Iterative-Standardized-Averaged-Uniform"</i>	Single time frame with 20 min duration (unless censored). The standardized, averaged images smoothed with a scanner-/ reconstruction-specific Gaussian kernel, to achieve a uniform isotropic resolution of $\sim 8^3$ mm FWHM (full-width-half-maximum). This is of special interest for PET analysis pipelines that include partial volume effect (PVE) correction, which usually request the entry of a filter parameter (which would be "8 mm" in the present case).
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As illustrated in Figure 4, existing differences in image appearance due to the varying total fields of view, voxel sizes and intrinsic resolution are incrementally reduced by preprocessing, with the final *Prep-4* version showing harmonized image quality:

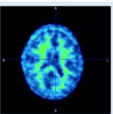
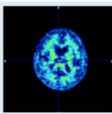
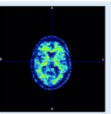
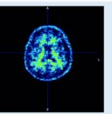
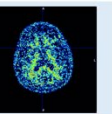
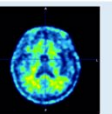
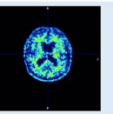
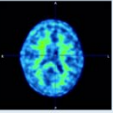
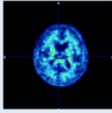
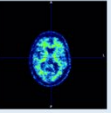
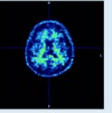
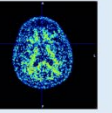
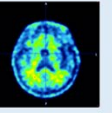
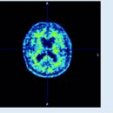
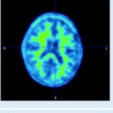
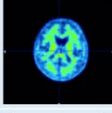
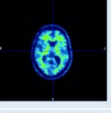
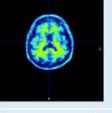
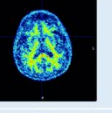
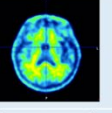
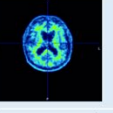
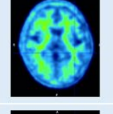
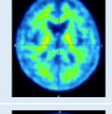
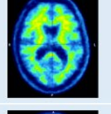
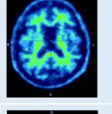
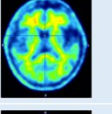
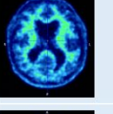
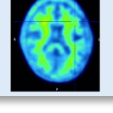
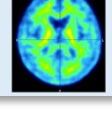
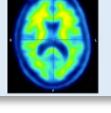
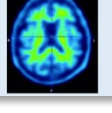
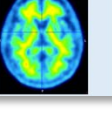
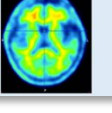
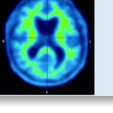
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Original							
Motion corrected							
Motion corrected + averaged							
Standardized + averaged							
Standardized + averaged + smoothed							

Figure 4. Schematic outline of the image preprocessing outcomes (Florbetaben examples)

4.5. DICOM data export

The original and preprocessed PET image data are both provided in **DICOM format** for consistency.

To harmonize file formatting, the original DICOM file names (which were highly variable between and even within sites) were replaced by a **unified naming scheme** *Tracer_SeriesDescription_Img_ImageNumber* (e.g. “FBB_Iterative_Original_Img_0007.dcm”) during final data export.

During data export, some additional **editing of DICOM header information** was necessary.

1. Most importantly, DICOM header addresses which could theoretically contain *location*- and especially *patient*-relevant information were deleted or replaced for **pseudonymization purposes**.

There were some exceptions for biological background parameters that are partially relevant for image interpretation: Birth year [YYYY0101, *if provided*], age, sex, body length [*if provided*], and body weight.

2. Since the original **DICOM series descriptions** varied widely for different vendors and scan protocols (which is different from the situation with the Siemens only MR protocols in DELCODE), they were replaced by **uniform Series Descriptions** (DICOM address: 0008,103E), also to simplify automation of later DICOM to Nifti file conversion and subsequent data processing.

Yet, the original series descriptions are written into the *Image Comments* field (DICOM address: 0020,4000) to keep the information traceable.

3. The Image Comments field also contains **supplementary information about reconstruction parameters** that are not reliably present in the regular DICOM addresses (e.g., number of iterations and subsets). For the Prep-4 series, this includes the additional Gaussian filters that were applied to harmonize image resolution.
4. PMOD changes or deletes some auxiliary DICOM addresses during image processing (especially: **vendor-specific private tags**). Therefore, relevant header information from the original DICOM series was rewritten into the PMOD-generated series during final data export.

This means the most reliable source of technical information remains the “Original” DICOM series.

5. Qualitative analysis of Florbetaben (FBB) amyloid positivity

To date, visual rating for cortical tracer binding by experience readers remains the clinical gold standard for evaluating amyloid PET images. Moreover, the qualitative evaluation of the regional tracer binding pattern is often still possible in situations where image acquisitions are suboptimal for quantitative analyses.

Accordingly, qualitative evaluations of amyloid positivity are available for a minority of DELCODE-PET datasets which had to be quarantined for other purposes.

5.1. Data preparation

To prepare for the qualitative ratings, the 4 x 5 min time frames of the original attenuation-corrected iterative reconstructions were motion corrected and averaged using PMOD.

5.2. Visual rating procedure

Qualitative ratings of amyloid positivity were based on **consensus judgments of two readers with clinical experience in nuclear medicine**.

Based on guidelines of the *NeuraceqTM Educational Programme for Healthcare Professionals* (provided by Life Molecular Imaging), qualitative rating procedures included:

1) **A priori evaluation of technical interpretability (yes/ limited/ no):**

General check for visible motion artifacts, suboptimal attenuation correction, atrophy, etc. Scans rated “no” will not be provided.

In general, even FBB scans which were quarantined for quantitative analyses, or DICOM data processing were still interpretable from the visual-qualitative perspective. Accordingly, the available number of qualitative FBB ratings is slightly higher than for the other PET data sources.

2) **Qualitative ratings, primarily based on the PET emission data.**

If necessary, complementary low-dose CT or MR image data were consulted (e.g., to evaluate potential influences of atrophic changes)

3) **Two-level qualitative rating of regional tracer binding**

Following established ratings procedures for Florbetaben (e.g. [Barthel et al., 2011](#)), visual evaluation of amyloid positivity was performed in a two-step procedure:

Step 1 Scoring of Regional Cortical Tracer Uptake (RCTU) for 4 predefined brain regions

RCTU	Meaning	Explanation
1	No Uptake	No cortical uptake in the majority of slices representing the brain region
2	Moderate Uptake	Smaller areas with cortical uptake, but in the majority of image slices representing the brain region
3	Pronounced Uptake	Large confluent area of cortical uptake, but in the majority of image slices representing the brain region

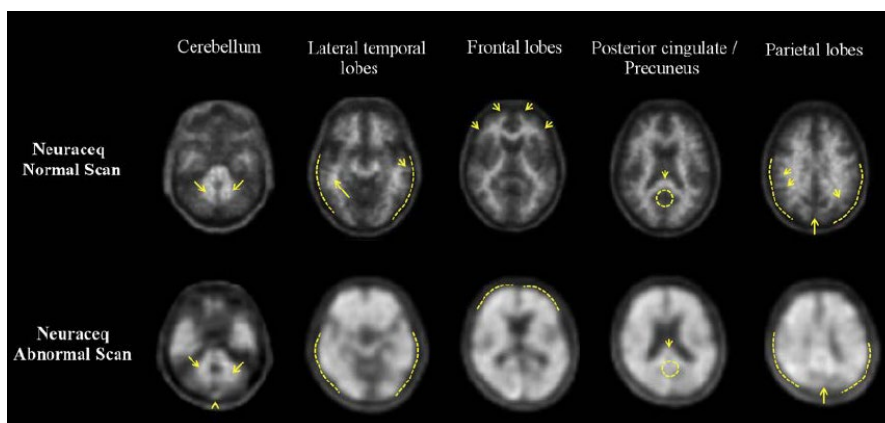


Figure 5. Brain regions for RCTU assessment ([European Medicines Agency](#))

Step 2: Assignment of Brain Amyloid Plaque Load (BAPL)

- Global evaluation of amyloid plaque burden
- Derived from previous RCTU scorings
- Three-level classification of beta-amyloid deposition
- Can be simplified to a positive/ negative categorization

Judgment	BAPL-Score		Conditions
Negative	1	No beta-amyloid deposition	RCTU = 1 for all 4 regions
Positive	2	Moderate beta-amyloid deposition	RCTU = 2 for at least one of 4 regions, but no region with RCTU=3
	3	Pronounced beta-amyloid deposition	RCTU = 3 for at least one of 4 brain regions

4) Evaluation of rating confidence:

Raters provided a judgment of their rating confidence, on a three-level scale: Low, moderate or strong.

5.3. Outcome variables for FBB qualitative analyses

Based on these procedures, the following main outcome variables for the FBB qualitative image analyses are provided:

FBB_QUAL_P0 FBB_QUAL_P24	<i>FBB Visual data interpretation</i> Binary classification of FBB PET: Amyloid-positive/ -negative
FBB_BAPL_P0 FBB_BAPL_P24	<i>FBB BAPL score:</i> 1- negative <i>Brain amyloid plaque load</i> 2- moderate 3 - pronounced
FBB_SURE_P0 FBB_SURE_P24	<i>Rating confidence:</i> 1= low 2=moderate 3=strong

6. FBB semi-quantitative analyses

In amyloid PET studies, **cortical amyloid burden** is frequently operationalized by **Standardized Uptake Value Ratios (SUVR)** from “late” phase emission scans, where tracer binding is supposed to have reached an equilibrium state ¹. SUVR measures provide a *semi*-quantitative estimate of cortical amyloid load, because cortical gray matter tracer uptake (=binding to amyloid depositions) is expressed relative to (=scaled to) a reference region which is assumed to show low tracer uptake.

While region-specific SUVR measures are generally possible, most studies use **global SUVR** values, i.e., aggregate tracer uptake values over a composite of cortical brain regions (for an excellent state-of-the-art review: [Pemberton et al, 2022](#)).

We provide **two alternative formats of global SUVR calculation**:

6.1. “Classical” SUVR (analogous to Barthel et al., 2011)

“**Barthel SUVR**” will refer to SUVR calculations that used similar aggregated cortical volumes of interest (VOI) and cerebellar cortical reference regions like the “classical” validation studies for the FBB tracer ([Barthel et al., 2011](#); [Sabri et al., 2015](#)).

These data analyses used the “*Iterative-Original*” PET image series described above, in native subject space.

Data analysis was **performed in PMOD** (PMOD Technologies LLC, Zurich, Switzerland), using the *Maximum Probability Atlas* processing pipeline implemented in PNEURO (Figure 6), analogous to the Florbetaben reference paper ([Barthel et al., 2011](#)).

Linear transformations were used for motion correction of the FBB time frames, and coregistration to the MPRAGE anatomical MRI image of the respective participant. The MPRAGE scan was segmented (using a routine with 6 tissue classes similar to “New Segment” in SPM12) to (a) derive normalization parameters (which allow non-linear warping of imaging data between patient and atlas template space), and (b) calculate individual tissue probability maps for gray matter (GM), white matter (WM) and cerebrospinal fluid (CSF).

These spatial transformations were used to inversely warp the AAL atlas template ([Tzourio-Mazoyer et al., 2002](#)) into the native PET space of the individual, and derive anatomical volumes of interest (VOI) definitions which were additionally masked inclusively with the GM

¹ For measures like distribution volume ratio (DVR) or binding potential (BPND), a dynamic scanning protocol with early dynamic emission scans would be necessary, which were not possible in the present study design.

probability map (default threshold: 50%), and exclusively with the CSF probability map (default threshold: 50%). To date, the processing pipeline does not yet include partial volume effect corrections.

For each of the delineated VOI, the average voxel value was calculated and extracted for further data processing. The selection of the anatomical regions was based on the original reference paper (H. Barthel, personal communication) to maximize compatibility with previous publications.

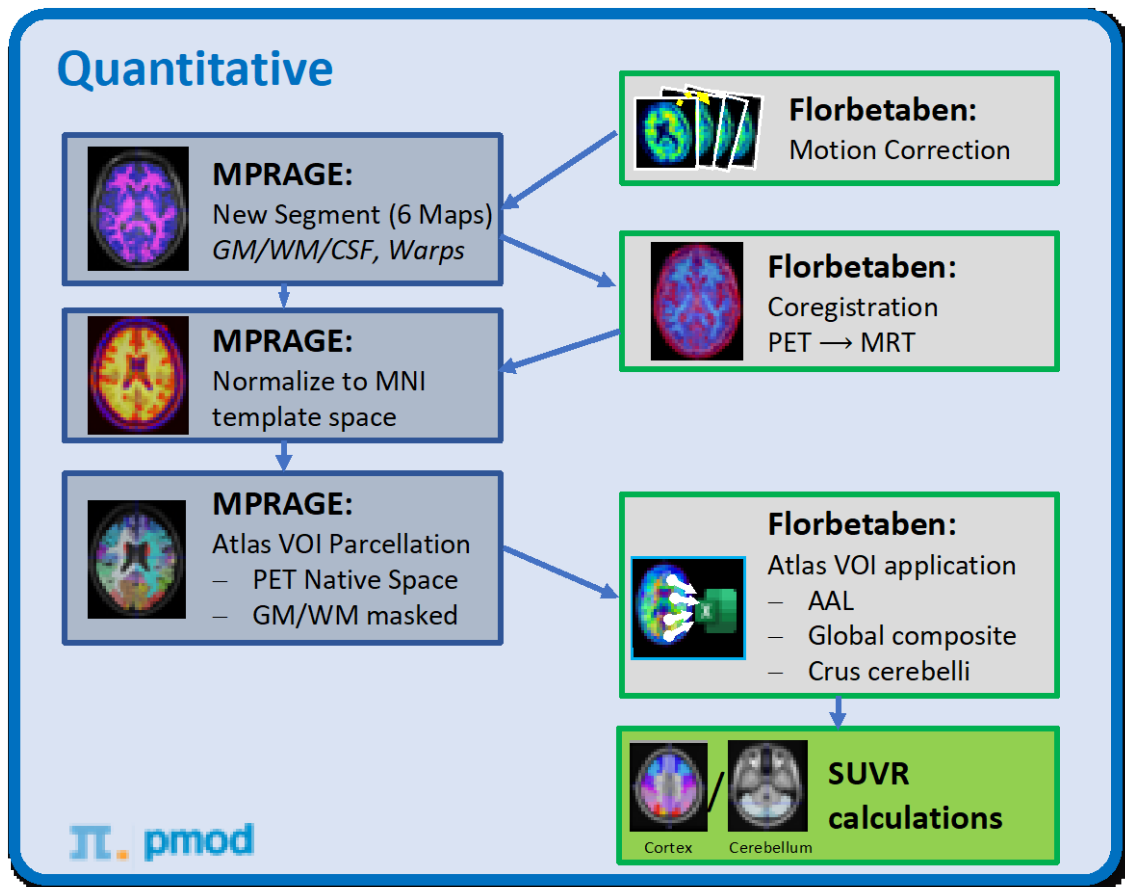


Figure 6. Florbetaben SUVR calculations with the PNEURO Maximum Probability Atlas pipeline

To examine cortical tracer uptake, a volume-weighted average of

- frontal,
- lateral temporal,
- parietal,
- occipital,
- anterior cingulate and
- posterior cingulate

anatomical VOI was calculated.

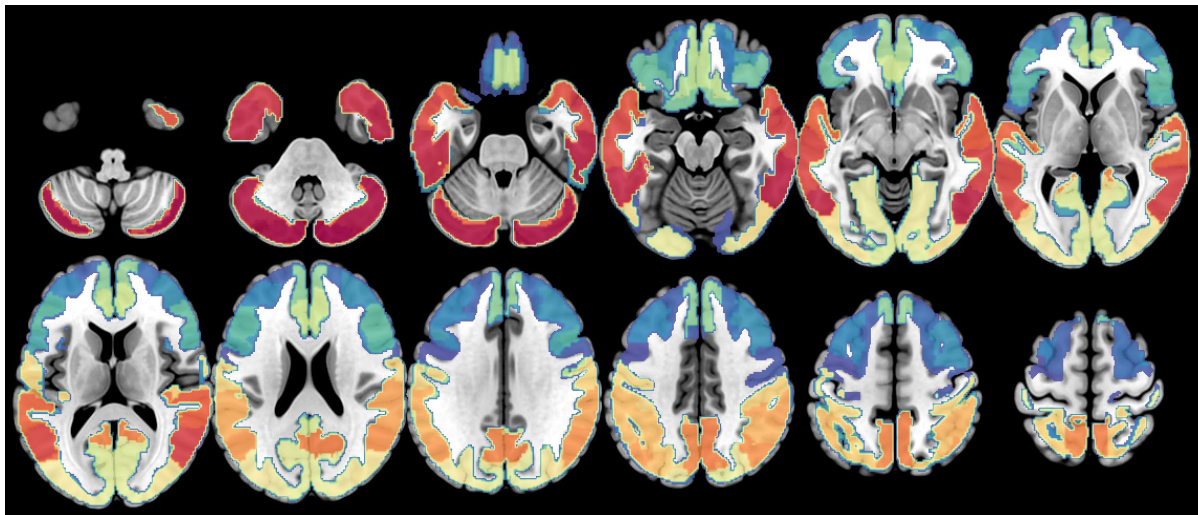


Figure 7: Anatomical regions in the AAL template used for delineation of global composite and cerebellar reference VOI, displayed on the MNI-152 template in MRICroGL. Note that the original AAL VOI definitions are actually larger, but get masked with the gray matter probability maps ($p > .5$) of each individual participant.

The value extracted from this global composite VOI was divided by the respective average value for the reference VOI (defined by the crus cerebelli of both hemispheres), to derive the SUVR score that represents the level of global cortical amyloid burden.

Note: The average activity uptake values for the multiple GM regions of the AAL atlas were extracted in PMOD, and aggregated off-line to derive the volume-weighted global composite VOI. While these regional values are currently *not* available for general release, there are technical options to generate tailored regional VOI composites for qualified research questions. In this case, please contact the PET group for further discussion.

6.2. Centiloid values

Because there are various tracers with slightly different characteristics, and various analytical strategies (software packages, brain atlases, target and reference region choices, etc.), there is huge interest to standardize amyloid load quantification and to make observations from different tracers and/or study cohorts more comparable.

The Centiloid protocol ([Klunk et al., 2015](#)) is an initial step into this direction: It does not only provide (1) a processing framework that – if performed appropriately – allows to produce Pittsburgh Compound B (PiB) SUVR calculations technically similar to the original study. Because the original study examined both young participants (=probably amyloid-negative) and clinical AD patients (=probably amyloid-positive), they could also (2) use the observed mean PiB SUVR of these two groups to scale the SUVR values, i.e., by defining “0” (=typical amyloid-negative) vs “100” (=typical amyloid-positive) reference points on a common, continuous “Centiloid” scale.

Note: This means that amyloid-negative individuals with SUVR levels below the young controls' *average* can have $CL < 0$, while amyloid-positive individuals with SUVR above the AD patients' *average* can have $CL > 100$.

While Centiloid was originally defined with PiB data ("Level-1 analysis" in [Klunk et al., 2015](#)), later studies collected PET data with both PiB and alternative amyloid tracers (including FBB: [Rowe et al., 2017](#)) *in the very same individuals*, and found that the SUVR values derived with the Centiloid pipeline were numerically different, but highly correlated between PiB and the alternative tracers. If sufficiently high, this allows regression-based linear transformations of the tracer SUVR to PiB SUVR *equivalents*, which can subsequently be transformed into Centiloid values ("Level-2 analysis" in [Klunk et al., 2015](#)). This way, an (approximately) common scaling of the SUVR values across different tracers becomes possible. Importantly, these pipelines can be adopted by other sites ("Level-3 analysis" in [Klunk et al., 2015](#)), if the given implementation of the analysis pipeline (e.g., in SPM, PMOD) does not introduce error (see below).

Data analysis was again **performed in PMOD**, using the PNEURO *Maximum Probability Atlas* processing pipeline with a modified version of the Centiloid atlas provided in the software. The processing steps and parameters followed the detailed description provided in an [Application Note by PMOD](#).

The main processing steps include:

- Preparatory motion correction and averaging of the original 4 PET time frames
- Automated cropping of the MRI and PET images
- Segmentation of the MRI image (using the SPM8 Unified Segmentation approach with three tissue classes for GM/WM/CSF)
- Coregistration of the PET image to the MRI image
- Normalization of the MRI and PET images (i.e., warping into MNI space)
- Delineation of the Centiloid atlas VOI for the cortical target region (CTX) and the whole cerebellum (WC) reference region in atlas (not: native) space (see: Figure 8)
- Extraction of mean voxel values for CTX and WC
- Calculation of the ratio: $^{Tracer}SUVR = CTX/WC$
- Computing of Centiloid values based on the pipeline-specific formula (see below):

$$CL = 154.472 * ^{Tracer}SUVR - 155.661$$

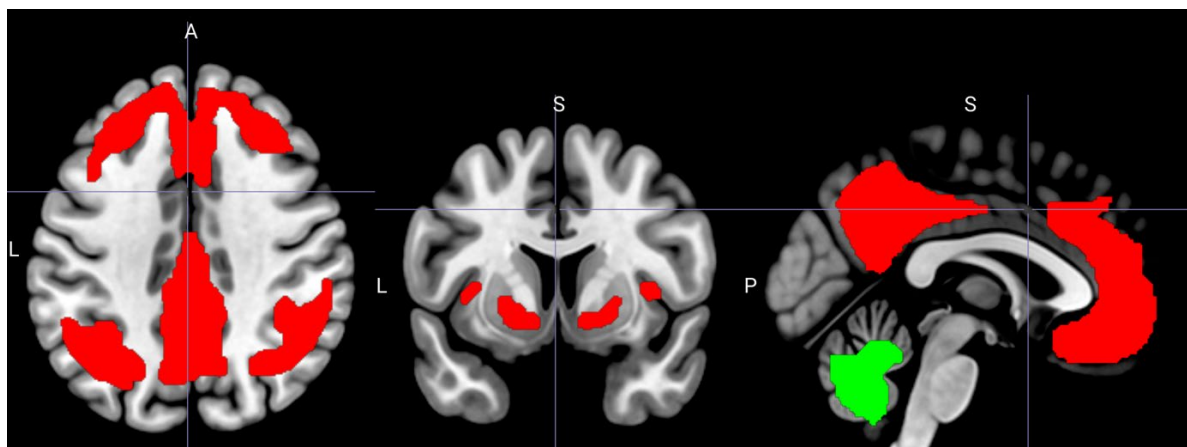


Figure 8: Cortical target (CTX) and Whole Cerebellum reference region (WC) used for Centiloid analysis, projected onto the MNI-152 brain template in MRICroGL².

Important: The applicability of the linear transformations depends on the assumption that data processing reproduces the original Centiloid processing pipeline with sufficient accuracy, especially since we did not use original SPM8, but an independent software.

To assure this, the Centiloid workgroup *demand*s empirical validation, which is based on a mandatory re-analysis of both the original PiB (Klunk et al. (2015)) and FBB datasets (Rowe et al., 2017), freely available from www.gaain.org/centiloid-project. The calculated SUVR and CL values have to be statistically compared with the original data analyses, and users may be asked to provide this benchmark information (e.g., in a supplement):

- “PiB (Klunk et al., 2015): PMOD mean $SUVR_{CTX/WC}$ values were $M(AD)=2.074$ and $M(YC)=1.011$ for AD patients and young controls, respectively, less than 2% different from the original results. Using these sample parameters for Centiloid scaling, regression of the PMOD-calculated Centiloid values with the original CL values fulfilled the established quality criteria (slope=0.999, intercept=-0.04, $R^2=0.999$).”
- “FBB (Rowe et al., 2017): PMOD mean $SUVR_{CTX/WC}$ values were $M(elderly)=1.43$ and $M(young)=1.01$ for elderly and young participants, respectively, less than 2% different from the original results. Regression of PMOD-calculated Centiloid values with the original CL values fulfilled the established quality criteria (slope=1.005, intercept=-1.103, $R^2=0.999$). For direct transformation of PMOD-calculated FBB SUVR into PMOD-calculated CL, linear regression provided the following equation: $^{FBB}CL = 154.472 * ^{FBB}SUVR - 155.661$.”

In general, it should be pointed out that the Centiloid approach only allows to achieve *approximate* comparability across tracers and study cohorts: For example, the original authors

² Note that the Centiloid target region is more restricted than the cortical region used in the Barthel SUVR calculation. Moreover, there is no gray matter masking of the VOI regions, and the reference region includes both gray and white matter of the cerebellum. Despite these technical differences, preliminary analyses of DELCODE baseline datasets suggest that the correlation between the two measures is quite high ($r=0.966$).

([Klunk et al., 2015](#); [Rowe et al., 2017](#)) note that potential technical influences of different PET camera systems, reconstruction protocols, etc. have not been investigated systematically yet (which is still valid). This may contribute to the observation that pooled data analyses (e.g. [Salvado et al., 2019](#)) make converging observations, but still find some inter-sample variation, e.g. regarding cut-off values for amyloid positivity (e.g. [Farrell et al., 2021](#)).

To reduce the potential effects of different scanner resolutions, we performed two Centiloid analysis for each dataset: An initial analysis based on the original “raw” PET image series (complementary with the “Barthel SUVR” analysis described above), was recently supplemented by an analysis of the preprocessed image series with harmonized voxel geometry image resolution of ~8 mm FWHM.

6.3. Outcome variables

The main variables are the semiquantitative parameters for global cortical tracer binding:

FBB_SUVR_P0 FBB_SUVR_P24	<i>FBB: Standard Uptake Value Ratio</i> Following Barthel et al. (2011): Relative to cerebellar crus
FBB_CL_P0 FBB_CL_P24	<i>FBB: Centiloid (original)</i> Based on the original image series
FBB_CL_8mm_P0 FBB_CL_8mm_P24	<i>FBB: Centiloid (8mm resolution)</i> Based on preprocessed FBB PET images with harmonized image resolution of 8 mm FWHM

7. Complementary variables

In addition to the PET data types of interest (i.e., images, or analysis results), descriptive variables are provided for each examination. These variables describe technical aspects of the PET examination (which may be requested by reviewers), but also quality-related issues.

FBB_*_COMM_P0 FDG_*_COMM_P0 FBB_*_COMM_P24 FDG_*_COMM_P24	<i>Quality comments</i> For each data type (DICOM, quantitative and qualitative analyses), a commentary variable is provided that contains information regarding limitations of usability, or recommendations for potential use, if necessary.
FBB_DATE_P0 FDG_DATE_P0 FBB_DATE_P24 FDG_DATE_P24	<i>Date of PET examination</i>
FBB_REF_P0 FDG_REF_P0 FBB_REF_P24 FDG_REF_P24	<i>Clinical reference visit</i> Especially for FBB, PET exam dates may fall in-between two consecutive DELCODE main study visits. Here, the study visit closer in time is suggested for reference (which can be the next one).
FBB_REFMR_P0 FDG_REFMR_P0 FBB_REFMR_P24 FDG_REFMR_P24	<i>MR reference visit (M0/ M12/ M24....)</i> Since clinical visits not always included an MR examination, or the latter was scheduled on a different date, the MR visit closest in time was not always identical with the clinical reference visit. Sometimes, no contemporary MR data were available, but at least from the year before and/ or after PET exam. Again, the MR visit closer in time is suggested (in some equivocal cases, both). If no recent (< ~1 year) MR was available, the field was coded "N/A". Without a recent MR, MR-based pipelines for PET analysis are compromised, and without any MR, not possible. Here, PET-only analysis pipelines which spatially normalize images with a modality-specific template are less flexible, but generally possible. While this is well-established for FDG (e.g. Della Rosa et al., 2014), complementary solutions for amyloid PET emerged more recently (e.g. rPOP: Iaccarino et al., 2022).

AGE_FBB_P0 AGE_FDG_P0 AGE_FBB_P24 AGE_FDG_P24	<i>Age at time of PET examination</i>
FBB_DOSE_P0 FDG_DOSE_P0 FBB_DOSE_P24 FDG_DOSE_P24	<i>Decay-corrected injected tracer activity</i> [in Megabequerel, MBq] For some participants, the injected FBB tracer dose was lower than defined in the DELCODE protocol (260-300 MBq), but still consistent with manufacturer's suggestions (300 ± 60 MBq) in most cases. Tracer over- and underdosings were not generally excluded, if visual image quality was normal, but this variable can be used for selecting participants more stringently. For DICOM users: Note that <i>Radionuclide Total Dose</i> (0018,1074) will often show slightly different values. The information is entered manually at the scanner console, and was frequently based on a simple subtraction of syringe activity before and after injection (which is valid only with short delays between the measurement, due to the exponential decay of tracer activity). Therefore, available checklist information was used to calculate more precise estimates.
FBB_DELAY_P0 FDG_DELAY_P0 FBB_DELAY_P24 FBB_DELAY_P24	<i>Time between tracer injection and start of PET emission</i> [in min] Based on available DICOM and checklist information. For some participants, emission scan did not start regularly, i.e., 40 (FDG) or 90 min (FBB) post-injection, usually due to technical problems. While there is preliminary empirical evidence (Barthel et al., 2017) that Florbetaben SUVR values are robust across various time windows between 45 and 130 minutes, users can use this variable to select cases more stringently. For DICOM users: Note that <i>Radiopharmaceutical Start Time</i> (0018,1072) is not always reliable, since this time point is usually entered manually, and potential subject to typos. While this influences the absolute pixelwise Standardized Uptake Value (SUV) values, it should have no effect on the usual analytical approaches in brain imaging, which scale against a reference region (=effectively work with SUV ratio scores).

FDG_DIET_P0 FDG_DIET_P24	(<u>FDG only</u>) <i>Duration of food abstinence</i> [in hours] Duration was sometimes not reported numerically, but ordinal information was given (e.g., >6 hours = the requested minimum). Meanwhile, this was considered acceptable, since food abstinence mainly served as prerequisite for a low blood glucose level.
FDG_BGL_P0 FDG_BGL_P24	(<u>FDG only</u>) <i>Blood glucose level</i> [in mg/dl] From the blood glucose measurement performed immediately before FDG injection.
FDG_REST_P0 FDG_REST_P24	(<u>FDG only</u>) <i>Rest period after injection</i> [in min] The requested <i>minimum</i> resting period was 20 min.
FBB_PETCAM_P0 FDG_PETCAM_P0 FBB_PETCAM_P24 FDG_PETCAM_P24	<i>Vendor and model of PET scanner</i> • GE Discovery 690: PET/CT • Philips Gemini TF 16 TOF: PET/CT (@two sites) • Siemens Biograph Duo: PET/CT • Siemens Biograph mCT 128 Flow Edge: PET/CT • Siemens Biograph mMR: PET/MRI (@two sites)
FBB_DEVICE_P0 FDG_DEVICE_P0 FBB_DEVICE_P24 FDG_DEVICE_P24	<i>PET scanner ID</i> For optional use as a statistical covariate. Not always identical with clinical site, as some sites used the same PET scanner. For DICOM users: Identical with <i>Device Serial Number</i> (0018,1000) in DICOM header. Because the original vendor information was already deleted by some PET sites before upload, and for pseudonymization purposes, each scanner received a new, arbitrary ID string (“00001” - “00007”).
FBB_RECON_P0 FDG_RECON_P0 FBB_RECON_P24 FDG_RECON_P24	<i>Original PET reconstruction algorithm</i> Users who received image data can also find the relevant information in the DICOM header (note also the extended <i>Image Comments</i> field).
FBB_GEO_P0 FDG_GEO_P0 FBB_GEO_P24 FDG_GEO_P24	<i>Original PET image geometry</i> Users who receive DICOM image data can also find the relevant information in the DICOM header (note also the extended <i>Image Comments</i> field).

FBB_RESL_P0 FDG_RESL_P0 FBB_RESL_P24 FDG_RESL_P24	<i>Original PET image resolution</i> [in mm] Gaussian filter full-width-half-maximum (FWHM) for the original image reconstruction. Based on estimates that were derived from 3D-Hoffman brain phantom measurements with these reconstruction protocols (see: Paragraph 4.3).
FBB_FWHM_P0 FDG_FWHM_P0 FBB_FWHM_P24 FDG_FWHM_P24	<i>Additional PET smoothing kernel</i> [in mm] Size of the additional Gaussian kernel applied to the <i>Prep-4</i> image series to approximate an effective image resolution of 8mm FWHM. Practically relevant only for DICOM users, and users of the Centiloid quantifications based on the harmonized image data (Paragraph 6.2).
FBB_MAXTR_P0 FDG_MAXTR_P0 FBB_MAXTR_P24 FDG_MAXTR_P24	<i>Maximum translation between frames in x, y or z direction</i> [in mm] To allow a more targeted evaluation of original PET time series where movement-related artefacts may play a role. Values are derived from the PMOD realignment processing step.

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